

TESSERACT v2 Forecast Improvement Platform

Goal #3 – Execution Blueprint

Mission

Build a Forecast Improvement Platform that compares the current TESSERACT forecasting baseline against new forecasting approaches, validation methods, metrics, governance rules, and engineering improvements.

The platform must provide evidence-based recommendations and a visible Shiny application that evolves after every completed stage.

Core Principles

1. Current TESSERACT is the baseline.
 2. New approaches must compete against the baseline.
 3. Data decides which approach is better.
 4. Shiny is read-only.
 5. Every stage must produce something visible.
 6. No stage is considered complete unless it leaves a visible artifact.
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STAGE 0 — Project Foundation

Objective

Create the local workspace and project structure.

Blocks

B0.1 Project Structure

Create:

```
tesseract-forecast-improvement/  
├─ data/  
├─ outputs/  
├─ scripts/
```

```
|— shiny_app/
|— docs/
|— notebooks/
└— tests/
```

B0.2 Environment Setup

- R
- Shiny
- Python
- Forecasting libraries

B0.3 Project Documentation

- README
- Roadmap
- Architecture Notes

Visible Output

Folder structure exists.

Completion Criteria

Shiny launches successfully from local machine.

STAGE 1 — Vision & Wireframe

Objective

Design the future platform before building it.

Blocks

B1.1 Executive Landing Page

Define:

- What improved?
- What worsened?
- What needs review?

B1.2 Tab Architecture

Design:

- Overview
- Forecast Overlay
- Accuracy Comparison
- Model Tournament
- Backtesting Explorer
- Governance

B1.3 Navigation Design

User flow between tabs.

B1.4 Executive Storytelling

How the dashboard explains results.

B1.5 UI Mockups

Wireframes.

Visible Output

```
docs/wireframes/
```

and

Shiny mockup with empty screens.

Completion Criteria

Entire navigation is approved.

STAGE 2 — Data Contract

Objective

Define the permanent interface between Python and Shiny.

Blocks

B2.1 Forecasts Schema

B2.2 Actuals Schema

B2.3 Metrics Schema

B2.4 Rankings Schema

B2.5 Run Metadata Schema

B2.6 Recommendations Schema

Visible Output

```
data/sample/  
  
forecasts.csv  
actuals.csv  
metrics.csv  
rankings.csv  
run_metadata.csv  
recommendations.csv
```

Completion Criteria

Shiny reads all schemas successfully.

STAGE 3 — Baseline Replication

Objective

Reproduce TESSERACT exactly.

Blocks

B3.1 Current Model Inventory

B3.2 Current Backtesting Logic

B3.3 Current Selection Logic

B3.4 Current Accuracy Metrics

B3.5 Baseline Validation

Production vs Replicated Output

Visible Output

```
outputs/baseline/
```

Shiny shows:

Actual vs Current Forecast

Completion Criteria

Replicated baseline matches production behavior.

STAGE 4 — Evaluation Platform

Objective

Create the analytical engine.

Blocks

B4.1 Accuracy Metrics

- MAPE
- wMAPE
- sMAPE
- MAD
- MPE

B4.2 Advanced Metrics

- RMSE
- MAE
- MASE
- Bias
- Drift
- Stability
- FVA

B4.3 Horizon Splitting

- Short
- Medium
- Long
- Strategic

B4.4 Ranking Engine

B4.5 Recommendation Engine

Visible Output

```
outputs/metrics/  
outputs/rankings/
```

Shiny Accuracy tab becomes functional.

Completion Criteria

Metrics calculated automatically.

STAGE 5 — Model Lab

Objective

Allow current and new models to compete.

Blocks

B5.1 Current Models

- ARIMA

- Prophet
- ETS
- Fixed Growth

B5.2 Statistical Candidates

- AutoARIMA
- Theta
- ETS Explicit
- Naive

B5.3 ML Candidates

- LightGBM
- XGBoost
- CatBoost

B5.4 Experimental Models

- N-BEATS
- N-HiTS
- TFT

Visible Output

```
outputs/model_lab/
```

Shiny displays:

Current vs Proposed

Completion Criteria

Model Tournament is operational.

STAGE 6 — Validation Lab & Governance

Objective

Improve forecast selection and validation.

Blocks

B6.1 Current Validation

B6.2 Sliding Window

B6.3 Expanding Window

B6.4 Out-of-Time Validation

B6.5 Horizon Validation

B6.6 Composite Score

B6.7 Recommendation Rules

- Keep
- Test
- Replace
- Review

Visible Output

outputs/governance/

Shiny Governance tab becomes operational.

Completion Criteria

Platform can generate recommendations.

STAGE 7 — UI/UX + LLM Insights

Objective

Transform technical outputs into business insights.

Blocks

B7.1 Executive Landing Page

B7.2 Interactive Cards

B7.3 Drill-down Navigation

B7.4 Forecast Storytelling

B7.5 Forecast Summary Generator

B7.6 Model Explanation Generator

B7.7 Recommendation Narrative

B7.8 Executive Weekly Summary

Visible Output

Shiny becomes executive-ready.

Completion Criteria

A manager can understand results in under 10 seconds.

STAGE 8 — Codebase Improvement

Objective

Close the engineering portion of Goal #3.

Blocks

B8.1 Notebook Consolidation

B8.2 Shared Utilities

B8.3 Centralized Configuration

B8.4 Output Standardization

B8.5 Structured Logging

B8.6 Governance Process

- PR Process
- CODEOWNERS
- Documentation

Visible Output

docs/codebase_improvement/

Completion Criteria

Formal engineering proposal completed.

Final Deliverables

Product Deliverables

- Shiny Platform
- Forecast Improvement Dashboard
- Model Tournament
- Governance Dashboard
- LLM Insights Layer

Technical Deliverables

- Evaluation Engine
- Data Contract
- Extended Model Library
- Validation Framework
- Codebase Improvement Proposal

Goal #3 Deliverables

- Ownership of TESSERACT forecasting architecture
 - Accuracy improvement analysis
 - Validation improvement analysis
 - Engineering improvement proposal
 - Visible prototype demonstrating improvements
 - Evidence-based recommendation framework
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Success Definition

At the end of the project:

1. Current TESSERACT forecasts are visible.
2. Candidate forecasts are visible.
3. Improvements are measurable.
4. Recommendations are automated.
5. Executives can understand results quickly.
6. Engineers can inspect details deeply.
7. The platform remains read-only.
8. Goal #3 is fully satisfied across accuracy, maintainability, automation, governance, and engineering process.